HW03-ICA :: Part A — Horses vs. Camels (Input Pipeline)

COSC 6373

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This notebook loads the dataset and builds a simple `tf.data` pipeline.

In [1]:
```
from __future__ import annotations

from pathlib import Path

import matplotlib.pyplot as plt
import tensorflow as tf

# CPU-only
try:
    tf.config.set_visible_devices([], "GPU")
except Exception:
    pass

print("TensorFlow:", tf.__version__)
print("Working directory:", Path.cwd())
```

```
TensorFlow: 2.16.1
Working directory: c:\Users\Adam\Desktop\projects_and_code\School\COSC6373\HW3\Part_A
```

In [2]:
```
# Dataset path
DATASET_DIR = Path("../dataset")
TRAIN_DIR = DATASET_DIR / "train"
TEST_DIR = DATASET_DIR / "test"

# Pipeline settings
IMG_SIZE = (224, 224)   # ResNet50 default input size
BATCH_SIZE = 32
SEED = 6373

print("TRAIN_DIR =", TRAIN_DIR.resolve())
print("TEST_DIR  =", TEST_DIR.resolve())
print("Train exists:", TRAIN_DIR.exists())
print("Test exists:", TEST_DIR.exists())
```

```
TRAIN_DIR = C:\Users\Adam\Desktop\projects_and_code\School\COSC6373\HW3\dataset\train
TEST_DIR  = C:\Users\Adam\Desktop\projects_and_code\School\COSC6373\HW3\dataset\test
Train exists: True
Test exists: True
```

In [3]:
```
def count_images(root: Path) -> int:
    exts = {".jpg", ".jpeg", ".png", ".bmp", ".gif", ".webp"}
    return sum(1 for p in root.rglob("*") if p.is_file() and p.suffix.lower() in exts)


if not TRAIN_DIR.exists():
    raise FileNotFoundError(f"Train folder not found: {TRAIN_DIR.resolve()}")
if not TEST_DIR.exists():
    raise FileNotFoundError(f"Test folder not found: {TEST_DIR.resolve()}")

print("Train images:", count_images(TRAIN_DIR))
print("Test images:", count_images(TEST_DIR))
```

```
Train images: 360
Test images: 40
```

In [4]:
```
train_ds = tf.keras.utils.image_dataset_from_directory(
    TRAIN_DIR,
    labels="inferred",
    label_mode="int",
    seed=SEED,
    image_size=IMG_SIZE,
    batch_size=BATCH_SIZE,
)

test_ds = tf.keras.utils.image_dataset_from_directory(
    TEST_DIR,
    labels="inferred",
    label_mode="int",
    seed=SEED,
    image_size=IMG_SIZE,
    batch_size=BATCH_SIZE,
)

class_names = train_ds.class_names
print("Class names:", class_names)
print("train_ds:", train_ds)
print("test_ds:", test_ds)
```

```
Found 360 files belonging to 2 classes.
Found 40 files belonging to 2 classes.
Class names: ['camel', 'horse']
train_ds: <_PrefetchDataset element_spec=(TensorSpec(shape=(None, 224, 224, 3), dtype=tf.float32, name=None), TensorSpec(shape=(None,), dtype=tf.int32, name=None))>
test_ds: <_PrefetchDataset element_spec=(TensorSpec(shape=(None, 224, 224, 3), dtype=tf.float32, name=None), TensorSpec(shape=(None,), dtype=tf.int32, name=None))>
```

In [5]:
```
print("train_ds element_spec:", train_ds.element_spec)
print("test_ds element_spec:", test_ds.element_spec)
```

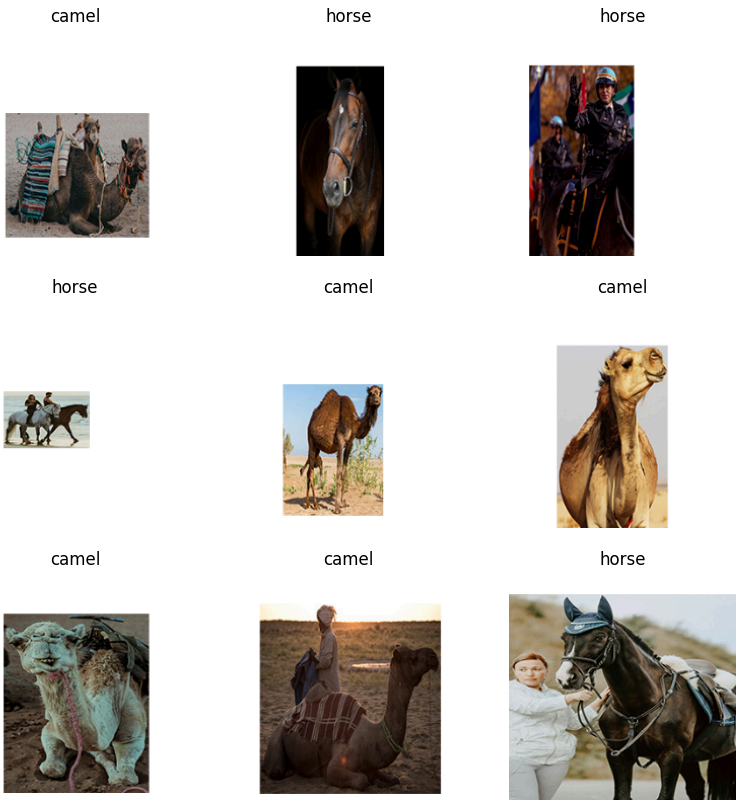
```
train_ds element_spec: (TensorSpec(shape=(None, 224, 224, 3), dtype=tf.float32, name=None), TensorSpec(shape=(None,), dtype=tf.int32, name=None))
test_ds element_spec: (TensorSpec(shape=(None, 224, 224, 3), dtype=tf.float32, name=None), TensorSpec(shape=(None,), dtype=tf.int32, name=None))
```

In [6]:
```
# Inspect one batch to confirm tensor shapes
images, labels = next(iter(train_ds))
print("images shape:", images.shape)   # (batch, H, W, C)
print("labels shape:", labels.shape)   # (batch,)
print("dtype:", images.dtype, labels.dtype)
```

```
images shape: (32, 224, 224, 3)
labels shape: (32,)
dtype: <dtype: 'float32'> <dtype: 'int32'>
```

In [7]:
```
# Visualize a few examples from the batch
plt.figure(figsize=(10, 10))
for i in range(min(9, images.shape[0])):
    ax = plt.subplot(3, 3, i + 1)
    plt.imshow(tf.cast(images[i], tf.uint8))
    plt.title(class_names[int(labels[i])])
    plt.axis("off")
plt.suptitle("Sample images from the input pipeline")
plt.show()
```

Sample images from the input pipeline



Notes

- `train_ds` and `test_ds` yield `(images, labels)` batches.
- `images` are resized to `IMG_SIZE` and have shape `(batch, H, W, 3)`.

Acknowledgment

GPT-5.2 was to help generate and organize parts of this notebook.

Specifically, cell 4 was revised using GPT-5.2